

NTP — Guided Notes ANSWER KEY

Unit 1.6.3 | CompTIA Network+ N10-009 Obj. 1.6 | N10-008 1.6

For instructor use / distribute after students complete the notes.

Question 1

List three network systems or services that break or behave incorrectly when clocks are not synchronized:

1. _____
2. _____
3. _____

Answer: Any three: Kerberos authentication (rejects tickets if clock skew exceeds 5 minutes); log correlation (events appear out of order across devices); certificate validation (wrong date causes rejection of valid certs or acceptance of expired ones); scheduled tasks/replication (fire at wrong time or collide).

Question 2

Stratum 0 is the _____ itself (e.g., atomic clock or GPS). Stratum 1 servers are directly connected to _____. Stratum 16 means _____.

Answer: Reference clock (hardware); a Stratum 0 source; unsynchronized — the server has lost contact with its upstream source.

Question 3

In a typical enterprise network, the internal NTP server syncs to an external _____ or _____ source. All other network devices then sync to _____. This design limits external NTP traffic to _____ connection.

Answer: Stratum 1; Stratum 2; the internal NTP server; a single.

Question 4

SNTP differs from NTP in that it _____. SNTP is acceptable for _____ devices. Full NTP is preferred for _____ where millisecond accuracy matters.

Answer: Simply snaps the clock to the server's time without continuously adjusting for latency/drift; client devices (workstations, phones, IoT); servers, routers, and infrastructure devices.

Question 5

NTP uses port _____ / protocol _____. Best practice is to configure at least _____ NTP sources so that clients can _____ responses and detect a bad time source.

Answer: UDP port 123; at least two; compare/validate.

Question 6

In a Windows Active Directory environment, workstations sync time to their _____. Domain controllers sync to the _____. The PDC emulator should be configured to sync to _____. If this last

step is skipped, the result is _____.

Answer: Authenticating domain controller; PDC emulator; an external NTP source; the entire domain's time drifts, eventually causing Kerberos authentication failures.

Question 7

A device reports Stratum 16. This means _____. To troubleshoot, you would _____.

Answer: The device is unsynchronized — it has lost contact with its upstream NTP source. Trace the NTP chain upstream: check if the configured NTP server is reachable (ping/UDP 123), check firewall rules for UDP 123, verify the upstream server itself is synchronized.

Question 8 — Real World Application

REAL WORLD: Monday morning, no staff can log in to their Windows workstations. Error messages say 'authentication failure' but credentials are correct. What is the first thing you check and why?

Answer: Check the time on the domain controller and compare it to an accurate external reference (e.g., time.google.com or the system clock on a phone). Kerberos authentication fails if client and DC clocks differ by more than 5 minutes. A drifted DC clock matches this exact symptom — all logins fail simultaneously with vague auth errors. Fix by correcting the DC clock and verifying NTP is configured and running.